

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		Gravitational (1) Earth (1)		2		2		
		(ii)		Equal and opposite Accept they are balanced	1			1		
		(iii)		Substitution: $1.5 = m \times 10$ (1) Mass = 0.15 [kg] (1)	1	1		2	2	
	(b)	(i)		10 [m/s²] accept 9.8 [m/s²]		1		1		
		(ii)		1.25 [N]		1		1		
		(iii)		Substitution into: $F = ma$ i.e. $1.25 \text{ ecf} = 0.15 \text{ ecf} \times a$ (1) Rearrangement: $a = \frac{1.25}{0.15}$ (1) $= 8[.3]$ [m/s²] (1)	1	1 1		3	3	
	(c)			No air resistance [on the moon] (1) Accept drag <u>So acceleration is same</u> so agree (1) Accept speeds up the same			2	2		
				Question 3 total	3	7	2	12	5	0